

LAN Pdf

1) Explain use of ping command in Network?

A) The ping command is one of the simplest yet most powerful tools in networking - it checks whether a device (host) is reachable, measures latency (delay) and detects packet loss, making it essential for troubleshooting connectivity issues.

2) Explain netbios protocol?

A) NetBIOS (Network Basic input/output system) is a legacy networking interface developed in the 1980s that allows computers in a Local Area Network (LAN) to communicate using names instead of IP addresses. It provides three main services: name resolution, session management, and datagram communication.

Although largely replaced by modern protocols like DNS and SMB over TCP/IP, NetBIOS is still present in Windows systems for backward compatibility.

3) Explain SMB protocol?

A) The SMB (Server Message Block) protocol is a network file-sharing protocol that allows computers to access files, printers, and other resources over a network. It's widely used in Windows environments but also supported by Linux and macOS, making it a cornerstone of resource sharing in both homes and enterprise.

4) Explain ping process from 50.0.0.10 to 50.0.0.20?

A)

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      1. Host
      2. icmp Echo Request
      3. ↓
      4. Source IP: 50.0.0.10 | Source MAC
      5. Destination IP: 50.0.0.20 | Destination MAC (after resolving ARP)

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50.0.0.20 Reply

Source IP: 50.0.0.20

Destination IP: 50.0.0.10